Movies suggestion with personalization using Matrix Factorization

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Authorship Declaration

I, Sai Bhone Myat Naing, confirm that this dissertation and the work presented in it are my own achievement.

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Abstract

Nowadays, there are thousands of online movie streaming websites, applications and platforms. However, some people are busy with their working times and cannot have enough time to search for movies or series in a short moment. Also, for some newbies who start to watch movies and do not have too much knowledge about movies.

In this study, we can learn one of the machine learning methods will reduce these kinds of problems. In the previous era, Content-Based and Traditional Collaborative filtering (Memory-Based) take place in every field such as products, movies, restaurants, services and so on. Nowadays, lots of different kinds of movies are increasing day by day. Consequently, the users' favorites could be different. Thus, the recommending system cannot recommend with only item features and similar users' favorites. That is why this recommender system has been applied. The appliances of this project are 1M Movie Lens Dataset, Matrix Factorization method (main method), Pandas and NumPy. To measure the prediction accuracy, the regression models such MSE, MAE, RMSE model will be applied for evaluation. The result of the proposed system can recommend the user accurately with personalization because of using the model-based collaborative filtering system named as Matrix factorization. Hopefully, the customer will get a better experience and the result of the project will deliver to the customer very well with the accurate prediction result because of the exploitation of the thoroughly researched knowledge.

Keywords: Recommender system, Collaborative Filtering System, Matrix Factorization (MF), Stochastic Gradient Descent (SGD), Mean Square Error (MSE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

Chapter 1

Chapter 1 contains of the introduction of this thesis, background, problem statements, aim & objectives, Scope and limitation and dissertation structure. Thus, we can know overview information this thesis by reading this chapter.

1.1. Introduction

There is an uncountable amount of products data on the Internet. Moreover, people likeness typically become more and more different. Thus, the recommender system plays important role in every field. People do not have too much extra time in their daily life, and they cannot waste their time on searching for something. Mostly, they get the information from the people recommendation. For instance, if the people want to know something about food, movies, hotel and so on, they ask their friends or the people they know about which one is good. However, they couldn’t get suggestions if their friends did not know it. To help humans in daily life, this system has been proposed.

1.2. Background of recommender system

In the past century, the smart devices usages were not widespread or these were not invented. Hence, people used to spend their free time on other kinds of entertaining stuff like reading, doing exercises and so on. Even if they spent their leisure time by watching movies, they would watch only on Television. Additionally, there are not used to have the Internet. Even though it used to have the Internet, the Internet fees were so expensive and people couldn’t afford to use it. Thus, most people only asked their friends about the movies’ taste and which movies do they prefer. Technologies become better and better as Times change. Nowadays, every people have at least one smart device such as a cell phone, tablet and so on. Anyone can use the Internet easily if he or she has a sim card and mobile phone. In addition, people can do whatever they like such as watching, posting, downloading, playing, communicating and so on by using internet.

In the past century, most people used to watch movies on Television and also with CDs, DVDs and so on because the Internet usage was not widespread. Thus, most people did not know that there are lots of movies over the world. In the Internet era, most people upload, download and watch movies on the Internet. Nowadays, more than millions of movies are available on the Internet. On the other hand, people nowadays must work more hours to withstand their lives rather than the people from the old century. In the old century, only one person had to work in one family and the whole family could withstand with that person’s income or salary. Nowadays, every family member except the old people have to work daily with more work time. Therefore, people have a little amount of time for relaxing. According to these reasons, people cannot watch and waste their relaxing time by searching movies on the Internet. For searching and watching movies as they wish in a short time, most top-tier movie platforms such as Netflix, Disney+, Amazon prime and so on use the recommender system to support and attract to the customers.

After Internet usage become widespread, people like shopping on the Internet rather than shopping in a shopping mall. There are also more than thousands of online sites for product selling on the Internet. On the other hand, the Covid pandemic has threatened people not to go out. Thus, purchasing products on the Internet become more and more. From a perspective of the marketing system, if the company could recommend or show related products that the customer bought or rated and then these products are also his favor, he may like the company service very much and may become loyal customers for this company. Thus, also Amazon, eBay and other kinds of online product seller sites use the recommender system with many techniques to recommend their products to the customer. The recommender system use recommendation not only for products but also for applications recommendation.

In the past, people use the recommender system with many programming languages such as java, C# and other programming language. Nowadays, some people might have used the AI method to apply on recommendation to get much better result. In AI, machine learning and deep learning recommender technique have been applied on the recommender system. The most well-known companies which used the AI recommender systems are Netflix, Amazon, Tinder, Facebook and YouTube (Drishya Bhattarari, 2021). All of this recommender systems except YouTube work with machine learning when YouTube uses deep learning.

YouTube uses Deep learning recommender system. Its performance is much better than machine learning recommender systems. Moreover, if we have Gmail account, we can upload our videos as our desires. Thus, millions of videos are increasing per second from unknown users. In addition, YouTube can also use for streaming. As a consequence, more than millions of videos and streaming videos must be recommended to the millions of users with their personalization. This is why, YouTube uses a powerful deep learning recommender system (Paul Covington et al., 2016).

Netflix uses content filtering and mainly Matrix factorization which is from collaborative filtering system to recommend movies for the user (Yehuda Koren et al., 2009). This Netflix recommending working process is similar as this proposed system. In 2006, Netflix announced the prize would be awarded to a group of people who could make 10% improve in the performance of movie recommendation system and the dataset used was from Netflix 100millions rating across 17000 movies by 500,000 anonymous users. Within 2 years, there was no one who can make this result (Yehuda Koren et al., 2009). “BellKor’s Pragmatic Chaos” won the Netflix prize in 2009 by using SVD (Singular Value Decomposition) (Mani Madhukar, 2014).

This dissertation’s target is ability to recommend the user accurately and to be better than the previous traditional recommender system likes Content Based Filtering and Neighborhood Collaborative filtering. But this proposed system does not have enough time and workload to implement a powerful deep learning recommender system. That is why, matrix factorization has been chosen as a main method which has more accurate prediction than content-based and memory-based collaborative filtering. This proposed system will help the daily life of people by reducing movie searching time.

1.3. Problem Statements

Most of the recommender system are implemented with Content based filtering system in past decade. This recommender system cannot recommend the user accurately because of loads of movies are overwhelming and people will differently favor. Developers mostly filter with Java and SQL language in the past. Even if it can suggest movies to users, it cannot be accurate in most of the conditions. Later, people use AI recommender in their system. Among AI recommender system, most of people use machine learning which has content-based and collaborative. For content-based filtering system, it can deal with new but cannot provide accurate data. In the past, even if they applied the collaborative filtering system, they would use only the memory-based system which has lots of limitations. Additionally, people’s favorites are changing day by day because of millions of movies are overwhelming on the Internet. Consequently, it is hard to know what the people actually like. If we do not use powerful recommendation system, we cannot provide the good experience to the user. These are the reasons or motivative facts which makes this dissertation has chosen to be implemented.

Matrix factorization has lots of model and types. Each model and type have different features and performance and different situation to use. Thus, picking a correct and an appropriate matrix factorization could be one of the problems. Similar matrix factorization implemented sites is not in real as much as it is supposed to be. The source of Matrix Factorization could be hard to find and explore. Matrix factorization uses Mathematical metrics to predict. To get a knowledge of machine learning field and know how to handle complicated datasets using with difficult metric, mathematics skill plays an important part.

1.4. Aims & Objectives

Aims

- Examine and evaluating Matrix Factorization algorithm.
- Construct the movie recommender with personalization by using matrix factorization method to recommend movie searcher.

Objectives

To achieve 1st aim

- Research, explore and analyze the related works of Machine learning based recommender system.
- Explore and evaluate up-to-date methodology which is collaborative filtering (Matrix Factorization) thoroughly for developing Movie recommendation system.

To achieve 2nd aim

- Design and implement proposed system with Python3.
- To evaluate and measure the predicted result with RMSE, MSE, MAE evaluation metric.

1.5. Scope and Limitation

In this dissertation, the project is mainly focus on matrix factorization recommendation. Firstly, Machine Learning will be learned about what is Machine Learning. After that, Content-based filtering will be explored about how does it work and the limitations. Memory based collaborative system will be explored to know the drawbacks and how it works. Matrix factorization which is the main methodology of this proposed system will be thoroughly explored. The time line of proposed dissertation starts from 22nd November and ends in 5th August which takes 185 days except weekends. This time could give a chance to get much better result. This proposed system will deliver for busy person, movie lovers, movie starter or newbie to be convenience and satisfied. Movie Lens data set will be used which has nearly 1millions of movies to make more choices for users. The knowledge that can get from this study of this dissertation are Machine Learning Recommending Algorithm and ability to handle the complicated data set and metrics.

Even though this project proposed time has nearly 9 months, these fully 9 months cannot be applied only on this project dissertation because other modules have to focus to finish in each deadline. On the other hand, there might be issued in time because of some threats such as health problems, personal case and other unexpected threats. Thus, most of project tasks processing time could be deviated. That time limitation could narrow the expected proposed system result. Time limitation might issue implementing time which can cause the website visuality will not be as well as expected GUI. Moreover, prediction accuracy might be lower than the expected accuracy result. Additionally, the research and exploring knowledge will not be as much as it will be supposed to get because of narrow researching time. In addition, the online website domain is expensive and it has to pay monthly. Moreover, cloud data server is one of the high-priced data servers. This is why this recommender system will be an offline recommender system because of financial limitation.

1.6. Structure of Dissertation

This dissertation composes 6 Chapters. All of the chapters are arranged by ordering every step of this proposed project.

Chapter 1 explains the introduction, background, aims and objectives and scope and limitation of this dissertation

Chapter 2 describes the theoretical background of the recommender system.

Chapter 3 provides literature review which contains the researching and overviewing information of the other work done.

Chapter 4 demonstrates Methodology and implementation. In Methodology, we can learn about models, pandas and datasets that are applied in this model demonstration. In implementation, we can learn every step of model demonstration.

Chapter 5 describes result and evaluation. It also provides discussion about the result and evaluation.

Chapter 6 explains future work and provides the conclusion of this thesis.

This is the end of the Chapter 1, we learned in this chapter about why this project has to be proposed, aims and objectives of projects and overview information of this thesis. The next chapter will be explained about the theoretical information of this proposed system.

Chapter 2

Before we start learning about the recommender system, we need to know what is AI and what is machine learning. This makes easy to study about the model. This chapter contains the theoretical background of this study and we can learn the theoretical information of AI, Recommender Systems and Evaluation models from these chapter.

2. Background Theory

2.1. Artificial Intelligence

In previous decade, number of computers usages are smaller than today decade. By increasing usage of computers, the technologies improve rapidly. Most of the top tier companies work mostly with robots in most of the advanced systems because they can task better than human. Nowadays, Artificial Intelligence usage become wider and wider in any field such as Education, Workplace, Cybersecurity, Healthcare and so on. Artificial Intelligence is the machine or system which imitates intelligence of human to make tasks and it can also improve based on the collected information by itself (OCI., (n.d.)). Only a few programming languages such as Python, R and Java are included and popular in Artificial Intelligence (Burns et al., 2022).

Main components of Artificial Intelligence are:

1. Machine Learning
2. Neural Network
3. Computer Vision
4. Cognitive Computing
5. Natural Language Processing
6. Speech Recognition

2.2. Machine Learning

Machine learning is the part of an Artificial Intelligence which provides system which automatically learning by itself without needing the explicit programmed (Expert.ai Team, 2021).

Machine learning has three types of algorithms:

1. Super-vised machine learning algorithm
2. Unsupervised machine learning algorithm
3. Semi supervised machine learning algorithm

2.2.1. Supervised machine learning algorithm

Supervised machine learning algorithm that is performed with labeled datasets to predict output values and to train the model (Pedamkar, 2021). The data which is tagged output of some data is called labelled data. Two main parts are classification and regression. Supervised machine learning is mostly used in Weather Forecasting, Products price prediction, traffic jam prediction and stock prediction (Daniel Johnson, 2021).

2.2.2. Unsupervised machine learning algorithm

Unsupervised machine learning algorithm that is performed with unlabeled datasets to act on that data no supervision. It makes more preferable than supervised machine learning because of unlabeled data is easier to get than labeled data (JavaTpoint, n.d.). Even though this algorithm can handle more complex processes than supervised machine learning, it cannot predict better than supervised learning.

2.2.3. Semi-supervised machine learning algorithm

Semi-supervised machine learning algorithm is the mixture of supervised and unsupervised machine learning which used both of the data which in huge amount of unlabeled and small amount of labeled data. As a consequence, semi-supervised machine learning used both benefits of algorithm to train data and predict more accurate result.

2.3. Deep Learning

In past decade, machine learning usage was popular in most of the fields. After improving the technology, human start to deploy deep learning which is more advanced and has more powerful in every performance than machine learning. Deep learning is the subset of machine learning which works with the brain function which is also called artificial neural network. Moreover, it can also deep learning which can handle large amount of training data and predicts more precisely than machine learning.

2.4. Recommendation system

Recommendation system is used to suggest the things to the user using with many different factors (Dwivedi, R., 2021). Therefore, most of well-known companies use recommender systems to improve customer relation management and to get a great opportunity. Not only in business but also in many other kinds of fields such as education, government and so on.

To receive the user’s preference, the system deals with a large volume of data and the information that is taken from user’s commitment to find out the match between items and user’s records using the similarity metrics. After getting items, system suggest these items to the user.

Types of recommendation systems are:

1.Popularity-Based Recommendation

2.Classification Model

3.Content-Based Recommendation System

4.Collaborative Filtering

2.4.1. Content-based Filtering

Content-based filtering is a kind of behavioral recommendation which used the implicit data or explicit feedback of the user. It processes with the features of the items to recommend the other similar features items to the users that was based on the user histories (Google Developer, 2021). Content-based filtering metric are cosine similarity and Jaccard similarity. Content-based filtering also has limitations such as Limited Analysis, Homogeneity and Cold Start Problem.

2.4.2. Collaborative filtering

Collaborative filtering system which is used with calculation metric and calculated the past data of the user selections and predicted by itself and recommend the user by comparing with predicted result. Collaborative filtering uses the explicit data of past user. Collaborative filtering system has two types of approach such as Neighborhood approach and Latent approach. Neighborhood (Memory Based) has two types of systems such as user-based and item-based recommender system. Model based or latent factor approach is Matrix Factorization. Collaborative filtering has some limitations such as Data sparsity, New User and New Items Problems.

2.4.3. Hybrid Filtering

Hybrid filtering system is the combination of both content-based system and collaborative filtering system. This mixture of two models helps to reduce the limitations of both of content-based filtering and collaborative filtering. There four ways to combine for hybrid filtering system. Implement both of content based and collaborative separately at first and then combining the prediction result way. Another way is content based recommendation features add into collaborative system and the other way collaborative features add into content-based system. The last way is building unifying model using both content-based and collaborative filtering system. This filtering system can provide more accuracy prediction than other approach because it uses both advantages of filtering system.

2.4.4. Matrix Factorization

Matrix factorization, is used for generating latent features while using multiplication of two matrix such as user and items. Rating can receive by the dot or multiplication of user matrix and item matrix.

$$\boxed{r_{ij} \approx \bar{u}_i \cdot \bar{v}_j}$$

Equation 1 Equation of Matrix Factorization (M. Tabari et al.,2018)

r_{ij} is the rating, $\bar{u}_i$ is matrix of user and $\bar{v}_j$ is the matrix of items. Moreover, figure 2 shows that 1st user rated Harry Potter, Shrek and Dark knight rises. Right side table shows ratings data. To give an example of how it is processed, we will use one of the adequate of records (ratings of user 1 on movie4). Matrix multiplication is row x column. So that, 1x1 and .1x.9 and then combine these two results. Then, it will get 1.09 which is equal with rating of movie 4 for user 1.

For user 1 prediction, he has already had user matrix and we need to predict ratings of movie 2 for user 1. However, movie 2 does not have item matrix. We can calculate for this movie 2 matrix by using other users' preference such as user 2 and 3 who rated this movie. Now, we get movie 2 matrix. Hence, we can easily predict rating of movie 2 for user 1. Also, user matrix of user 2 can be calculated by using the movies matrix that she rated. Therefore, all of the matrix of users and items and ratings depend on all users' preference or ratings records. That is why, it is named as latent factor model.

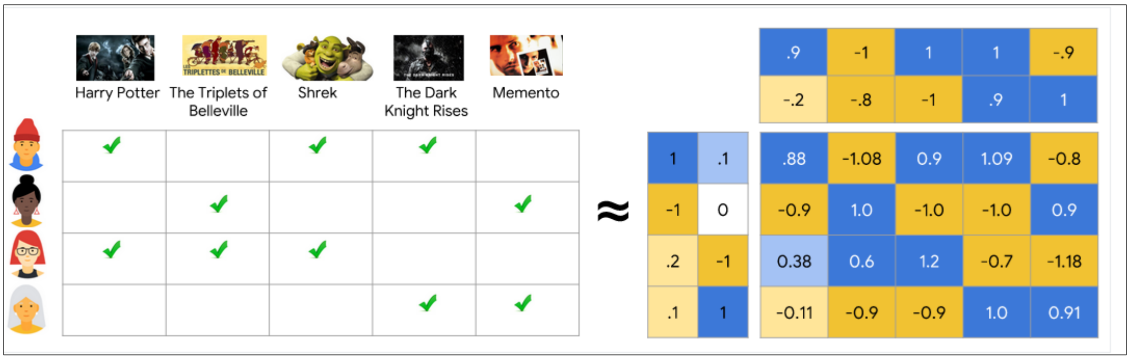

Figure 1 Matrix Factorization Process (Google developers, 2022)

2.4.5. Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent is one of the assistance algorithms to find the best fits between the actual and predicted results. Most of implementing Matrix Factorization are applied with SGD for minimizing the loss. This is metric which is based on Matrix factorization.

$$\boxed{\begin{aligned} q_i &\leftarrow q_i + \gamma \cdot (e_{ui} \cdot p_u - \lambda \cdot q_i) \\ p_u &\leftarrow p_u + \gamma \cdot (e_{ui} \cdot q_i - \lambda \cdot p_u) \end{aligned}}$$

Equation 2 minimizing equation of SGD for matrix q and p (Yehuda Koren et al., 2009)

$$\boxed{e_{ui} = r_{ui} - q_i^T p_u}$$

Equation 3 Error equation of SGD (Yehuda Koren et al., 2009)

q_i is the matrix of items, p_u is the matrix of users, γ is learning rate, λ is the regularization and e_{ui} is the error which is got from subtraction of the real rating r_{ui} and $q_i^T p_u$ which is multiplication or dot of user or item matrix or matrix factorization model prediction. First equation is to get new estimate q_i matrix of items. Second equation is to get new estimate p_u matrix of users. Learning rate γ has two types such as machine learnable parameter and hyper parameter. Machine learnable learning rate is an algorithm that perform itself for training of a particular datasets Hyper parameter learning rate is used as a variable for optimization of the machine learning algorithm. The learning rate in this equation is hyper parameter. Regularization acts as bias for avoiding overfitting model.

2.4.6. Alternating Least Square (ALS)

Alternating least square is one of the minimizing models of collaborative filtering system. It used problems of least square to minimize error throughout the time of update lines of actions that are used on associated vectors of the matrix's factors. W_i is weight or user matrix, H_j is height or item matrix, $r_{i,j}$ raiting or predicted result, λ is the regularization and I is matrix identity.

$$\boxed{\begin{aligned} W_i &= \sum_{r_{i,j} \in R_{i,*} \;\; and \;\; r_{i,j} \neq 0} (r_{i,j} H_j)/(H_j \cdot H_j^T + \lambda I) \\ H_j &= \sum_{r_{i,j} \in R_{*,j} \;\; and \;\; r_{i,j} \neq 0} (r_{i,j} W_i)/(W_i \cdot W_i^T + \lambda I) \end{aligned}}$$

Equation 4 Minimizing Equation of ALS for weight and height (Ömer Faruk Aktulum ,2019)

2.5. Evaluation Methods

Evaluation method are used for measuring the quality of demonstrated or researched Machine learning model. Perfect model result is very difficult to receive. That's why evaluation methods to measure to reduce the model error. There are lots of metrics to test the models.

- Classification
- Regression
- Confusion Matrix
- Logarithm loss
- Other metrics.

2.5.1. MSE (Mean Square Error)

MSE which is also known as MSD (Mean Square Deviation) the standard of the evaluation method to measure the deviated value in the models of statistical. The big value of MSE result means that errors of model is much. The lesser value, the better predicted value.

$$\boxed{MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

Equation 5 MSE (Study, n.d.)

y_i = Actual Result

$\hat{y}_i$ = Predicted Result

n = number of data

2.5.2. RMSE (Root Mean Square Error)

RMSE which is also known as RMSD (Root Mean Square Deviation) the standard of the evaluation method to calculate the prediction amount of model errors. It is also the root version of Mean Square Error. Moreover, it is also used to measure the deviation between actual result and predictive result.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}$$

Equation 6 RMSE (James Moody, 2021)

y_i = Actual Result

$\hat{y}_i$ = Predicted Result

n = Number of Data

2.5.3. MAE (Mean Absolute Error)

MAE which is also known as MAD (Mean Absolute Deviation). This method uses to measure the average distance between mean of the data and data points (predicted value). Comparing to MSE, MSE uses square of subtracted data and MAE uses positive subtracted data.

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n}|y_j - \hat{y}_j|$$

Equation 7 MAE (Stephanie, 2020)

y_i = Actual Result

$\hat{y}_i$ = Predicted Result

n = Number of Data

We knew about the models. In next chapter, we will study existing works before the demonstration of proposed model.

Chapter 3

Before we learn methodology or implementation stage, we will research and review other papers or existing works to receive a great source to expand the quality of implementation stage. This chapter contains the review of the literature about the recommendation system.

3. Literature Review

Good researching and overviewing other work done or existing works make a good dissertation result. There are loads of papers or articles about recommendation with Machine Learning and Deep Learning on the Internet. We will research and review about some papers about Machine Learning recommendations in this article.

Recommend system is one of the service provider systems by supporting the users in choosing with user preference's items or products with the ways of handling an enormous load of online data. Several recommender systems have been demonstrated and used since 1990-mid. Most of the researchers approved that recommender systems can provide a great opportunity in business, education, government and other fields (Jeffrey Lund et al., 2017). Recommendation system is used not only in entertaining webapp but also other fields such education, government, e-commerce and so on. Suja C. M. et al. (2021) demonstrated two prototypes of machine learning recommendation model to use on bank service and movie recommendation systems. Moreover, Abdon Carrera Rivera et al. (2018) researched mapping study with recommender systems. Mapping study is researching the academic papers from the online sites. The percentage of mapping study result which is regenerated with recommender systems are Hybrid 46%, CF 30%, Content 2%, Knowledge based 2% and other approach got 20%. Hybrid got the highest percentage rate. This means that recommender systems becoming one of the assistances in education field. Additionally, M.E. Cortes-Cedial et al. (2017) took survey and researched to improve the e-governance with recommender systems. They discussed how to applied the recommender systems in every case of e-governance. This means recommender system brings many advantages in every field.

Incoming era, most of the developer uses AI such as Machine learning or Deep Learning to get the most accurate personalization. Machine learning has two types of recommendation content-based and collaborative filtering. These two algorithms are explained in section 2.5.1 and 2.5.2. However, Content-based and collaborative based has some kinds of challenges and limitations like Data Sparsity, Scalability and Cold Start Problem. Mani Madhukar (2014) and JamesLe et al. (2019_ explained detail information of limitations of content-based and collaborative filtering system. To overcome these limitations, hybrid filtering was invented by that use both of the advantages of content-based and collaborative filtering. Hybrid filtering uses both of implicit and explicit data. Implicit means the user's history, watched and other mouse-clicked action. Explicit data means the data which needs users to provide rating or some kinds of action likes buying, picking or so on.

Noor Ifada et al. (2020) compared difference between collaborative filtering and hybrid filtering. He used 100K Movie Lens Dataset and as for algorithms he chose K Mean clustering for collaborative and weighted for hybrid filtering. Overall results show that Collaborative filtering performance is obviously better than hybrid filtering. This means that even though hybrid filtering can overcome the cons of the collaborative, the prediction performance is still lack. This thesis targets to get better accuracy.

Deep learning system is a powerful recommendation model. Paul Covington et al. (2016) demonstrated YouTube by using deep learning model for candidate generation and ranking. Consequently, deep learning can handle advanced platform. The platform about YouTube is explained section 1.2 (6^{th} paragraph). Jeffrey Lund et al. (2017) demonstrated autoencoder with Movie Lens 25M datasets in streaming to compare with most of the recommender systems recommendation by splitting 90% training 10% validation data and evaluating with RMSE. Deep learning got the best result among these systems. One of the amazing results is that he proved that deep learning is 10 ten times better than K Nearest Neighbor (KNN) model. This makes deep learning system to be deliberately chosen. But there are still limitations. It takes load of times and loads of advance knowledge to implement. Some of issues of this dissertation such as time and difficulty, deep learning is not good choice to implement. One of collaborative filtering system (Matrix Factorization) is selected. This model can also provide better predicted result.

To study Matrix Factorization, we mustn't forget about Netflix prize competition. Because the people who won that competition with Matrix Factorization. Yehuda Koren et al. (2009) wrote about Netflix recommender system. After demonstrating, he found that Matrix factorization is mostly suited with explicit feedback such as ratings. In demonstrating Collaborative Filtering recommender system, we can choose two options ALS and SGD. SGD is mostly used for optimizing the explicit feedback using recommender system. For Alternating Least

Square, most of people use for reducing cold start problems to be able to perform in implicit datasets conditions. This means that implementing for Matrix Factorization which is mainly process with explicit feedback, SGD is more appropriate than ALS.

Even though the system does not have explicit feedback matrix factorization can also work with implicit feedback. M. Tang et al. (2018) proved dealing with implicit feedbacks by combining weighted AMAN (wAMAN) and Matrix factorization to upgrade the OCCF (One Class Collaborative Filtering) system for applying it in emotion-aware recommender system.

JamesLe et al. (2019) demonstrated two kinds of experiments. The first experiment is neural matrix factorization. The first experiment is demonstration of seven variant models of Matrix Factorization such as vanilla MF, MF with bias, MF with sidefeatures, MF with temportal effects, MF with mixture tastes, Factorization with machine model and variational MF. Vanilla MF is standard MF so that its equation dot of matrix of user and item. MF with bias gets by add bias as regularization into standard MF for solving missing data problems. MF with side features is by adding sociodemographic or implicit ratings of users into MF with bias. MF with temporal effects is for to be able to handle reflectively constant changes such as user's preference and item popularity. FM is a big machine of MF which is built to handle large and complex data which contains a large number of user interactions like tag or categorical data. MF with mixture taste is mainly processed with implicit data and item tastes likes genres. Variational MF is used with Bayesian methods to predict items ratings. Figure 3 shows evaluation results of models which are demonstrated with 1M Movie Lens. Models are model 1 (vanilla), 2 (bias), 3 (side features), 4 (temporal effects), 5 (Factorization Machine),6 (mixture taste) and 7 (variational).

Models	Train MSE	Test Accuracy
Model1	0.748	0.835
Model2	0.670	0.787
Model3	0.619	0.788
Model4	0.760	0.793
Model5	0.658	0.821
Model6	0.609	0.817
Model7	1.762	0.836

Equation 8 Performance of 7 variant models of Matrix Factorization

(James Le et al., 2019)

There are some ways to improve the predicted result. Weijie Wang et al. (2018) demonstrated rounding recommendation. Rounding technique that the predicted rating is rounded to the integer which is nearest. They used 100K Movie Lens Dataset to test for rounding and normal recommendation. They used some of the collaborative filtering such as MF and KNN. They measured the error with MAE and RMSE. Rounding recommendation result has better result. Rounding will be one of the quality-improving technique for recommendation system.

In addition, Rahul et al. (2016) demonstrated the combination of typicality-based Collaborative filtering and Matrix factorization calculation to solve the limitation of the traditional method of Collaborative filtering memory-based system. This combined model performance is obviously better than traditional collaborative performance.

All of this research results could be supportive to incoming chapter 4 which is about methodology. As for the researched result, vanilla MF will be selected which is simple to implement and has a good accuracy. For error handling, SGD could be a good choice to process with explicit feedbacks to train vanilla matrix factorization model. For the data selection, choose 1M which is not too big or too small. This dataset could be a good choice to implement with efficient amount of data.

Chapter 4

We studied information of Matrix Factorization from the existing works. We will apply using these researched sources in implementation stage usefully. This chapter contains information of the components that are applied on implementation and implementation phase.

4. Methodologies

4.1. Appliances

Firstly, we must study the information of appliances before we implement our model.

4.1.1. Pandas

Pandas is a python library which is one of the most popular and powerful tools of data sciences to use for data wrangling or munging and analysis (Admond Lee, 2018). Moreover, Pandas is an open source that was implemented by Wes McKinney (Bronshtein, 2019). It creates rows and columns of data frame in python by reading the CSV, SQL database or other kinds of data. It provides several methods and functions for data analytic. Most of data science demonstrators used Pandas for their data implementation and analytics.

In this model training, Pandas performs reading dataset and converting into array and lists for model training. There is no error or missing values in data for training model. So, data cleaning is not needed. Before demonstrating model, we used Pandas to analyze the data to know some information by filtering with some conditions or preferences.

4.1.2. NumPy

NumPy, is a powerful library to perform operations of mathematics and logic, also known as numerical python, contains multi-dimensional array and array processing tools, invented by Travis Oliphant in 2005. Moreover, it is able to process with fourier transform, matrices and linear algebra (W3School, n.d).It plays a vital role in handling the array or lists. Its performance is faster than traditional python libraries because NumPy contains locality of reference. Locality of reference means arrays of NumPy are stored at one continuous place which provides efficient access and manipulation arrays for the processes. NumPy is a heart of machine learning model demonstration.

In this model training, NumPy performs as a main function. Data of items and users are stored as arrays of NumPy for Matrix Factorization training. The training

NumPy features which are applied in this model training are .dot (multiplication), .array (converting data into array format), .zeros (creating 0 arrays for assigning data array) , .matrix (another converting array format function) , .arange() (generating range of data)) . .random.binomial (perform process with probability [count or numbers of trial , probability, size]) and .random.rand (creating random matrix).

4.1.3. Matplotlib

Matplotlib, one of most popular and efficient visualization tools for analyzing data. Using NumPy library to visualize the graph of results the machine learning model or evaluation models. It consists of numerous plots such as bar, pie, line, histogram, scatter etc.

In this data analysis, Pyplot has selected. Pyplot, which is one of the submodules of matplotlib, is applied for visualizing the results of model implementing or the conditions of datasets.

4.1.4. Experiment of Dataset

To build movie recommendation, we downloaded the raw data of 1M movie-lens datasets from GroupsLens. This dataset is released in Feb, 2003. Movies from this dataset are between 1995 and 2003. This dataset has three files. According to the README.txt of 1M datasets which has provided from groupslens, they said that most of data had filled by hand so that there might be accidentally caused data issues. So that the data must be cleaned to be able to process with the proposed system.

Table 1 Raw Datasets of 1M Movie Lens

Source File	Information	Attributes
Movies.dat	Movies.dat contains the movies information. This contains 18 types of genres such as action, adventure, comedy and so on. Count of total movies are 3952.	Movie ID, Title, Genre.
Users.dat	Users.dat contains users' information. 20 types of occupation are included. The names of occupation have shown in the README.txt. Total users are 6040.	User ID, age, gender, occupation, zip code.

Ratings.dat	Ratings.dat contains the ratings information given by the users. The ratings range that are given by users is from 1 to 5 with intervals of 1. This data contains userid and moviesid two foreign keys from movies.dat and users.dat. This dataset contains 1M ratings from 6040 users across 3952 movies. Each user has at least 20 ratings which are based on 5 stars scale methods.	User ID, Movie ID, Rating and timestamp.

Table 2 Cleaning Process of raw datasets for training Matrix Factorization

Data Source	Issues	Methods
Movies-cleaned.dat	1. After opening with data-cleaning app (openrefine), some of genre are not in the same entities so that extra entities are existed. 2. Movies title contain ':' which can cause data reading issues. Because we read that data with ':' separation. If movies titles contain ':', there will be column issues in movies data reading. 3. After analyzing the movies id, wrong indexing has found. This could lead wrong length of movies data.	1. Set in all of the movie's genres in a same entity and remove extra entities. 2. Replace with "-" to make sure not to be causing data reading error. 3. After correcting index numbers or movies_id, we found that 200-300 movies were missed to reached 3952 movies. The model cannot process if the movies data counts are not the same with the data information README.txt because users in ratings data process with 1-3952 movies_id. Thus, add some new movies for model demonstration.

Ratings-cleaned.dat	1. Even though ratings data is able to read, extra entities existed between each entity. This makes the data to be caused error in contributing the lengths of users and items for MF model.	1. Remove extra columns and change the separator into 'tab'.
Users-cleaned.dat	1. Data is readable but still has some issues such as extra columns existed and numeric occupations.	1. Do as the same action for extra entities. Numeric occupations do not interrupt in data reading but to be more efficient to use change them into the names of occupations according to the README.txt.

4.3. Proposed system design

In our proposed system demonstration, data splitting stage is starter stage. After finishing data splitting, three types of data are received. Test data used for loss functions. For error handling, both train and valid data are applied. Matrix factorization in this error handling is used mainly for model error producing to analyze. SGD is for minimize the error and loss function is producing loss of data for analyzing to perform SGD. After finishing of error handling, enhanced or improved matrix of users and items have received. These optimized matrixes are both applied on recommendation and also evaluation result. In recommendation, it will predict with enhanced matrix by using matrix factorization. Also in evaluation stage, these improved matrixes are used. Start to end is the user interface stage or recommendation stage. User needs to put top number and his or her id to get predicted result.

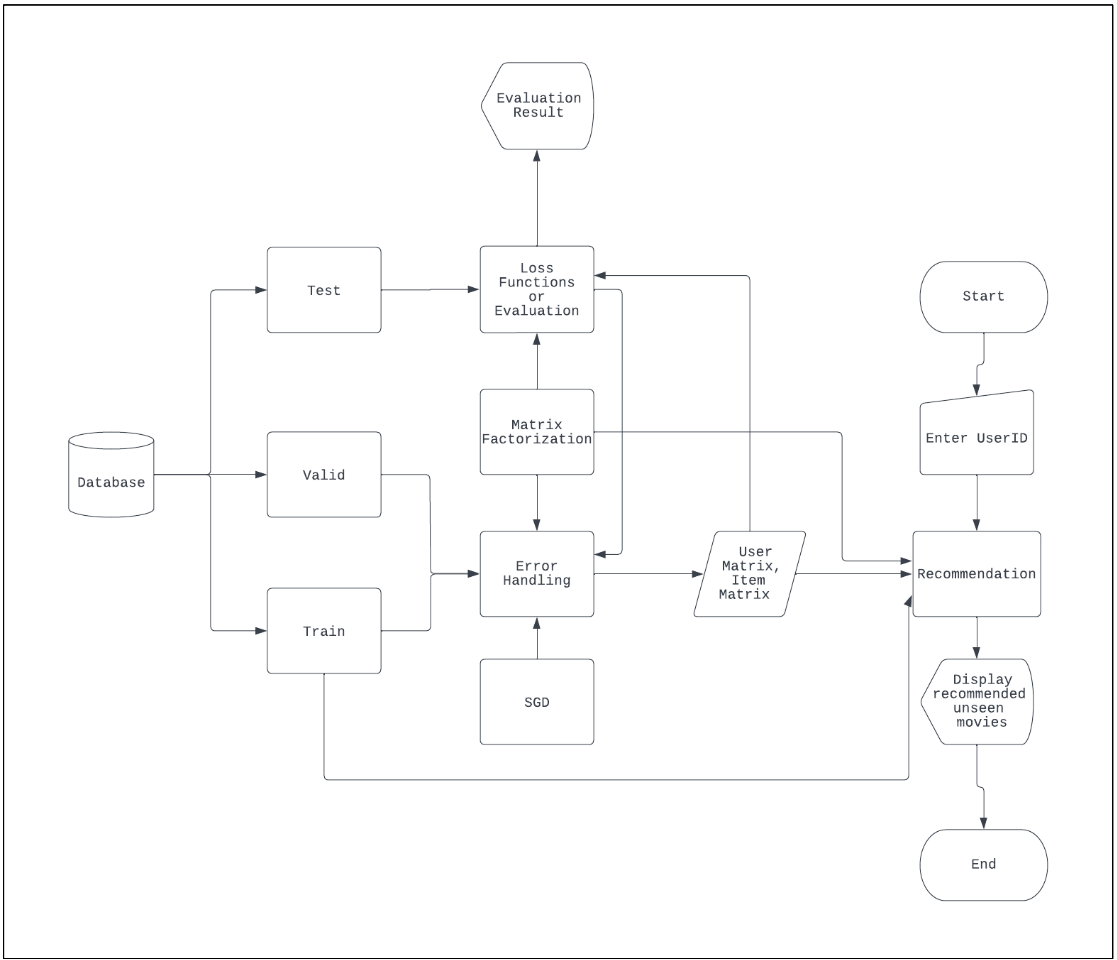

Figure 2 Proposed System Design

4.4. Implementation

Implementation has 6 stages. 1st stage is data analysis, 2nd stage is Data Splitting, 3rd stage is Matrix Factorization, 4th stage is evaluation or loss function, 5th stage is SGD, 6th stage is error handling and the 7th is recommendation stage.

4.4.1. Data Analysis

Before we start our implementation, we need to know our dataset situation. After that, we can know that the steps for training proposed model. In addition, we can also know that which library and components, functions, methods that we must apply and demonstrate.

Firstly, we must know who are our main target audience so that we must analyze our data the users record with occupation. This figure shows the data analysis of the movies watched depends on the users' occupations. Among this we found that most of the people who watched movies are students and the least is farmer. According to this result, new movies for adding in the data should be student's favor to get more movies watchers.

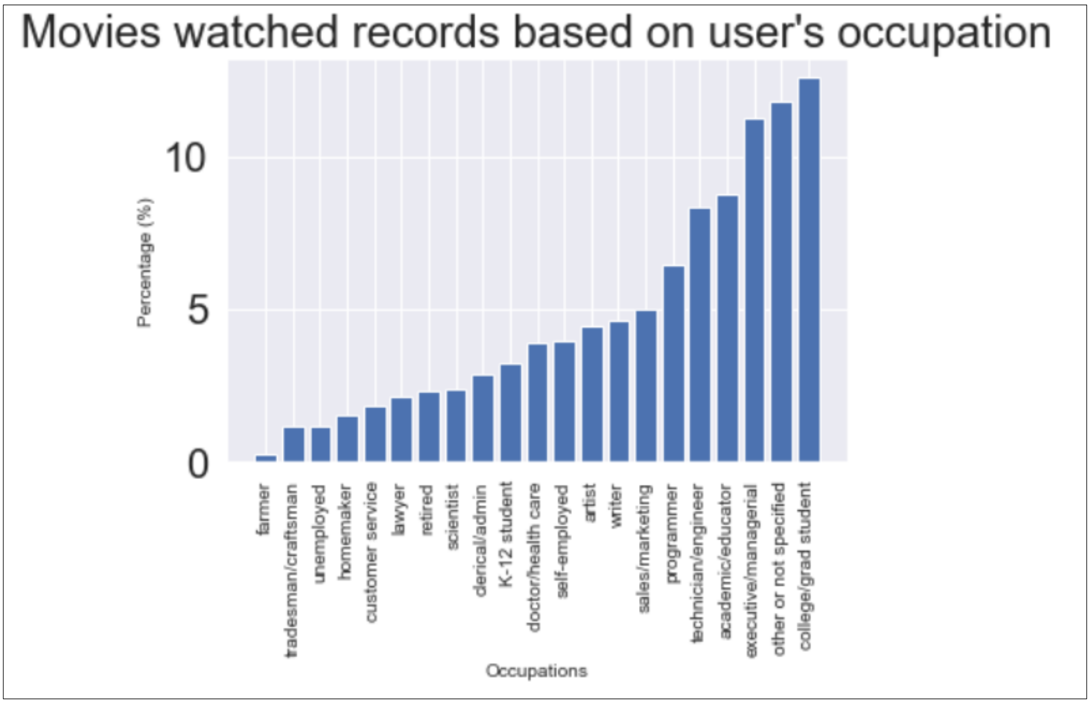

Figure 3 Data Analysis of movies watched data by user's occupation

Another data-analysis result we must know that is the situation of the relationship between movies and users. Figure 4 shows overall ratings records of movies from users. We can overview the numbers of ratings for each movie but cannot be visible about the detail information of the dataset clearly. Therefore, we will learn with another data analysis to know more information about the data.

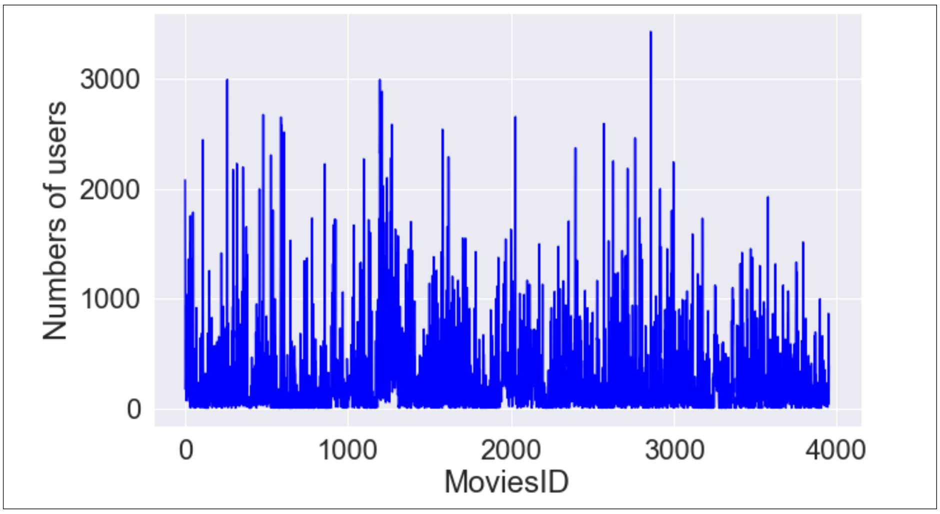

Figure 4 Data Analysis of every movies ID with numbers of users that are voted

Figure 5 shows the viewed records of numbers of movies depends on users. By view this chart, a few numbers of users watched more than 1000 movies but, a large number of users watched only 1%-2% of movies from this dataset. This means that most of users still did not know about or make action on most of the movies. Hence, we must deliver unseen movies to each user using their rating records. All of these data-analysis lead that delivering unwatched movies to each user will be one of the important output goals. We will start proposed model demonstration after knowing the target by data-analysis.

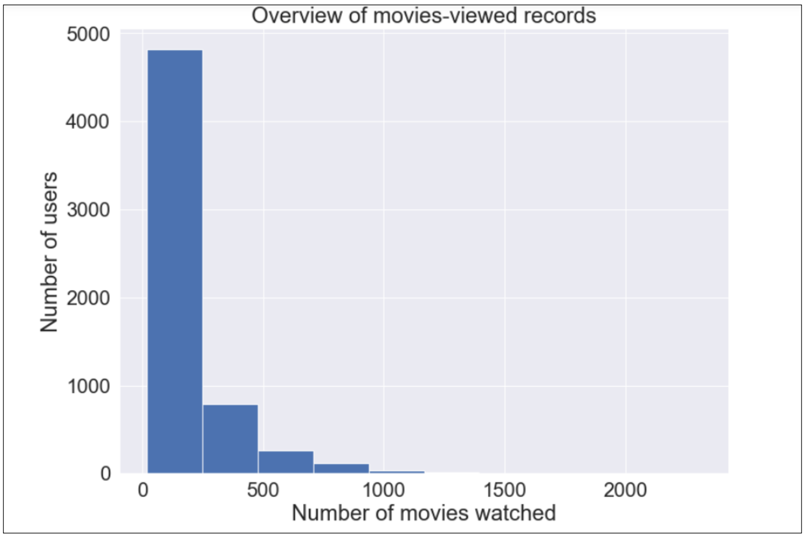

Figure 5 Data Analysis of users with numbers of movies watched

4.4.2. Data splitting

After finishing data analysis, we learned that most of users have already watched or rated only a few numbers of movies. Thus, we need to deliver to those users. The reason of data splitting is our model cannot be perfect so that we need to know our model performance. Additionally, most of MF demonstration applied with SGD model. Therefore, we need to split our data for analyzing the optimization result. Data splitting into three parts such as train, test and valid. Train set (60%) is for model training, model error handling and recommendation, valid set (20%) is for model error handling and testing set (20%) for evaluation results.

Firstly, read datasets as array for Matrix Factorization model appliance. Data converting function is used to pick the only rating number from the data to build new rating array format for each user. In figure 6, we can see one user rating record and this user rated ratings for movie id 1 is 5 and it is zero in second column by reason of he or she did not rate or seen this movie.

```
[5.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 3.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 4.0, 0.0, 0.0, 0.0, 0.0, 4.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.
0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.
0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.
0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0,
```

Figure 6 One of user rating records which are converted

After changing data, split the ratingscleaned.dat into training data (80%) and testing data (20%). Then split the 80% of training data with numpy.random.binomial function into 60% (train) and 20% (valid) of total dataset. Consequently, train has two set training set and validation set.

```python
def dataconverting(data, numbers_of_users, numbers_of_movies ):
    newdata = []
    for userid in range(1, numbers_of_users + 1):
        moviesid = data[:,1][data[:,0] == userid]
        ratingsid = data[:,2][data[:,0] == userid]
        ratings = np.zeros(numbers_of_movies)
        ratings[moviesid - 1] = ratingsid
        newdata.append(list(ratings))
    return newdata


#Number of Users and Movies
numbers_of_movies = len(moviesdata)
print(numbers_of_movies)
numbers_of_users = len(usersdata)
print(numbers_of_users)


training_dataset, testing_dataset = train_test_split(modeltrainingratingsdata, test_size=0.2)
con_trainset = dataconverting(training_dataset,numbers_of_users, numbers_of_movies)
con_testset = dataconverting(testing_dataset,numbers_of_users, numbers_of_movies)
def split_train_and_valid(data, ratio):
    training_set = np.zeros((len(data), len(data[0]))).tolist()
    validation_set = np.zeros((len(data), len(data[0]))).tolist()

    for i in range(len(data)):
        for j in range(len(data[i])):
            if data[i][j] > 0:
                if np.random.binomial(1, ratio, 1):
                    training_set[i][j] = data[i][j]
                else:
                    validation_set[i][j] = data[i][j]

    return [training_set, validation_set]

"""Spliting data"""
train = split_train_and_valid(con_trainset, 0.6)
test = con_testset
```

Figure 7 Process of Data Splitting

4.4.3. Matrix Factorization

For matrix factorization implementation, use one of the NumPy library features .dot function is applied which mostly used for multiplication of 2D arrays. P is matrix of users and Q is matrix of items. u and i is the IDs of users and items. The function output is matrix factorization model predicted results for users. This matrix factorization is vanilla matrix factorization so that models function needs only multiplication of matrix of users and items.

```python
def Matrix_Factorization(P, Q, u, i):

    return np.dot(P[u,:], Q[i,:])
```

Figure 8 Function of Matrix Factorization (Python)

4.4.4. Loss Function or Evaluation

To measure the Matrix Factorization model training performance, some of evaluation methods will be applied. MSE, RMSE and MAE, which are popular in measuring error of the recommendation models, have chosen for loss function or evaluation.

Figure 9 shows the function for calculation of MSE evaluation method. MSE equation is mean of power of actual data – predicted data. Number of users and items have already added at data converting state in session 4.4.2. It will calculate and combine the results of every user id. After that, it will be divided by evaluation count which depends on numbers of users and numbers of items.

```python
def MSE(data, P, Q):

    sum_of_error = 0.
    evaluation_count = 0
    numbers_of_users = len(data)
    numbers_of_items = len(data[0])

    for u in range(numbers_of_users):
        for i in range(numbers_of_items):
            if data[u][i] > 0:
                sum_of_error += pow(data[u][i] - Matrix_Factorization(P, Q, u, i), 2)
                evaluation_count += 1
    MSE = sum_of_error/evaluation_count
    return MSE
```

Figure 9 Function of MSE(Python)

Figure 10 shows calculation of RMSE evaluation method. The difference between MSE and RMSE is only square root. RMSE is the square root version of MSE so we need to just square root √ to get RMSE results. To get RMSE results, math.sqrt function is applied.

```
def RMSE(data, P, Q):

    sum_of_error = 0.
    evaluation_count = 0
    numbers_of_users = len(data)
    numbers_of_items = len(data[0])

    for u in range(numbers_of_users):
        for i in range(numbers_of_items):

            if data[u][i] > 0:
                sum_of_error += pow(data[u][i] - Matrix_Factorization(P, Q, u, i), 2)
                evaluation_count += 1
    RMSE = maths.sqrt(sum_of_error/evaluation_count)
    return RMSE
```

Figure 10 Function of RMSE(Python)

MAE evaluation calculation is shown in figure 11. By comparing with MSE, MAE does not need power. It means that the calculation is the mean of the subtraction of actual data and predicted data. But the calculation must be negative data because of without power. Negative evaluation is unacceptable. Consequently, we need to change positive result. To get positive data, abs() function is used for converting to absolute number or positive number.

```
def MAE(data, P, Q):

    sum_of_error = 0.
    evaluation_count = 0
    numbers_of_users = len(data)
    numbers_of_items = len(data[0])

    for u in range(numbers_of_users):
        for i in range(numbers_of_items):

            if data[u][i] > 0:
                sum_of_error += abs(data[u][i] - Matrix_Factorization(P, Q, u, i))
                evaluation_count += 1

    MAE = sum_of_error/evaluation_count
    return MAE
```

Figure 11 Function of MAE(Python)

Among loss functions or evaluation, layer 2 or MSE has chosen to regularize for avoiding from overfitting model and for getting the least model loss.

4.4.5. Stochastic Gradient Model

For model optimization, SGD model is chosen to minimize error. According to SGD model which has explained in session 4.1.2, we require the variables to perform SGD model such as error, P, Q, userid, itemid, features, learningrate and regularization. Corresponding with error metric which has shown in session 4.1.2, the error will get by subtraction of actual data and prediction (matrix factorization model). P and Q is matrix of users and items, features mean numbers of latent variable. This following is SGD model function with code:

```
def sgd(error, prediction, userid, itemid, features, lr, weigth_decay):


    for f in range(features):
        P[userid, f] = P[userid, f] + lr * (Q[itemid, f] * error - weigth_decay * P[userid, f])
        Q[itemid, f] = Q[itemid, f] + lr * (P[userid, f] * error - weigth_decay * Q[itemid, f])

    return P, Q
```

Figure 12 Function of SGD(Python)

4.4.6. Error Handling

This stage process uses loss function and SGD model to get better matrix of users and items for recommending function. In this the whole stage, SGD is mainly used for providing new matrix user and items which have already reduced error or loss. For analysis of loss function result, we will use statical graph for measuring how data is lost while we optimized with SGD model.

Figure 13 shows the function of error handling. The purpose this function is for producing better user and item matrix. The required components to perform the error handling function are training and validation data, features (numbers of latent), epochs, weight_decay(deep learning term) or regularization (machine learning term), lr (learning rate) and stopping (stop point for minimizing).

The required functions are Matrix Factorization, SGD and MSE. In this function, Matrix Factorization is used in two parts. The first part is producing the deviating point. The main function in this error handling is SGD. SGD model performs as minimizing model to get better model by using deviating point which got from subtraction of original data and prediction data, Matrix Factorization, features or numbers of latent, weight_decay or regularization and. Another sub-function is MSE. MSE performs with new optimized SGD to produce loss data of train and valid.

The whole error handling working process is SGD will minimize loss until using all of numbers of users and items in one epoch. After finishing minimizing with an epoch, it will get matrix items and users. MSE will use this new matrix items and users to perform loss function to produce loss of train and valid. Among this, if the subtraction of loss validation last array and second last array is less than stop point, this model will be assumed as the final optimized matrix of users and items. If not, it will continue the process of SGD to minimize loss to get the better model or matrix until it reaches its point. Also, epoch numbers will increase until it reaches its stop point.

```
def errorhandling(data, features=10, lr=0.0002, epochs=101, weigth_decay=0.02, stopping=0.001):

    train = data[0]
    valid = data[1]
    numbers_of_users = len(train)
    numbers_of_items = len(train[0])
    loss_train = []
    loss_valid = []
    P = np.random.rand(numbers_of_users, features) * 0.1
    Q = np.random.rand(numbers_of_items, features) * 0.1

    for e in range(epochs):
        for u in range(numbers_of_users):
            for i in range(numbers_of_items):

                if train[u][i] > 0:
                    error_ui = train[u][i] - Matrix_Factorization(P, Q, u, i)
                    P, Q = sgd(error_ui, P, Q, u, i, features, lr, weigth_decay)


        loss_train.append(MSE(train, P, Q))
        loss_valid.append(MSE(valid, P, Q))

        if e > 1:
            if abs(loss_valid[-1] - loss_valid[-2]) < stopping:
                break

    return P, Q, loss_train, loss_valid
```

Figure 13 Error Handling Process with SGD and loss function (MSE)

If the model has more loss data than normal rate. It is needed to change more appropriate setting to receive better matrix users and items for recommendation by analyzing the loss data. This action does not make model to be improved obviously. However, it can help to get a fit model. The better model, the more accurate recommended results will be delivered. Moreover, the curve forms of the train and validation can also depend on the setting for optimization even though the dataset is usable. By learning two curves distance or forms, we can know our datasets condition, requirements or optimization settings. Generally, the curves bad situation is divided into two types such as underfitting and overfitting.

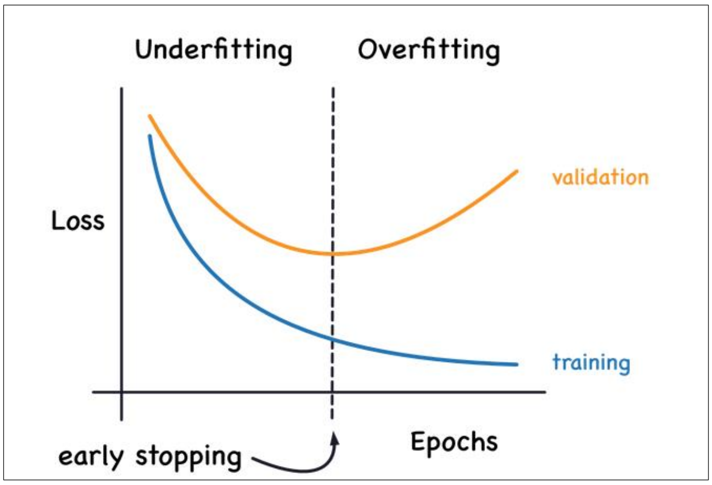

Figure 14 The curve situation that cause underfitting and overfitting (Ryanholbrook, 2022)

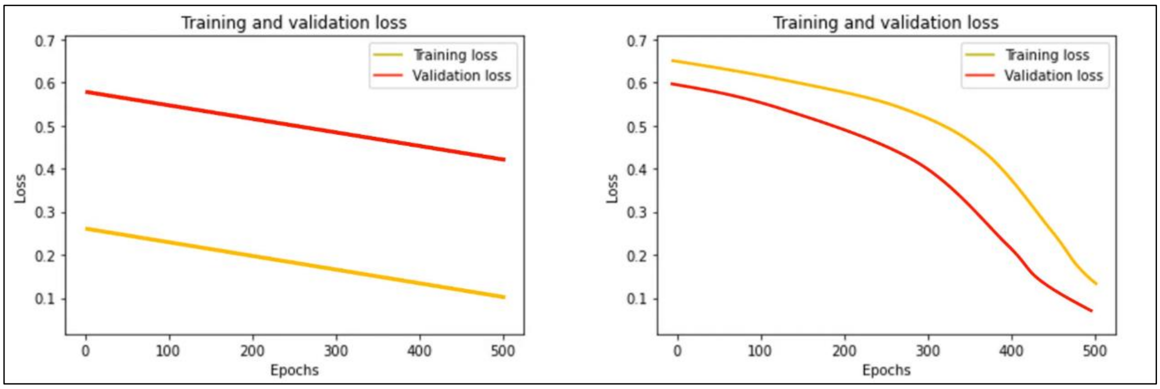

Figure 15 Sample graphs of underfitting (DigitalSreeni,2020)

Underfitting occurs when a machine learning model cannot capture the relationship of variables of input and output or the data underlying trend. Figure 15 shows model is in underfitting situation. This means data situation can't use for model training or learning. It can cause problem to the model accuracy which performs with testing dataset or less data. It can also cause problem in generalization for new data. Generalization is one of the abilities of machine learning model which can be able to perform well unseen or new data. To fit this situation, we need to add more epochs or to change the data sets because data is so complex to train model or to increase the data amount or to increase features or to reduce the stopping point for longer process duration.

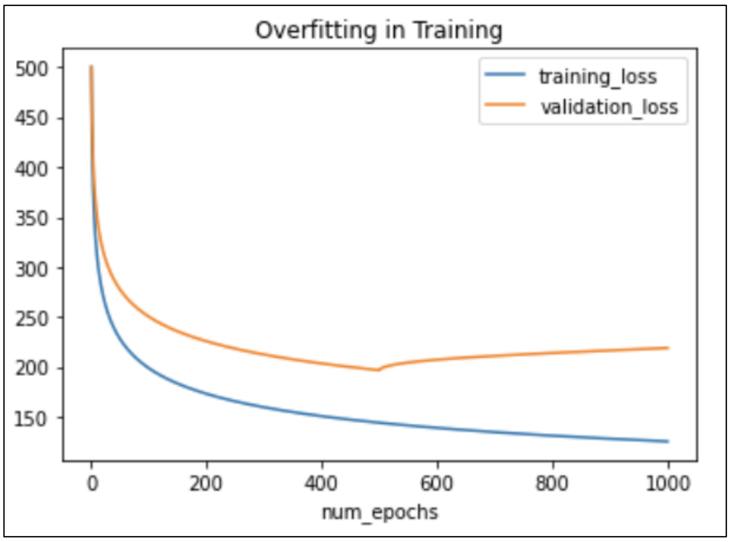

Figure 16 Sample graph of overfitting (Jacob Solawetz, 2020)

Overfitting occurs when machine learning model cannot perform well in evaluation. Figure 16 shows model training is in overfitting situation. This is cause by validation loss curve is too high and the gaps between training loss and validation loss is too far. To solve overfitting, we need to generalize the model. For generalizing, features have to be changed or add the appropriate regularization or increase stopping point or add more data into training dataset.

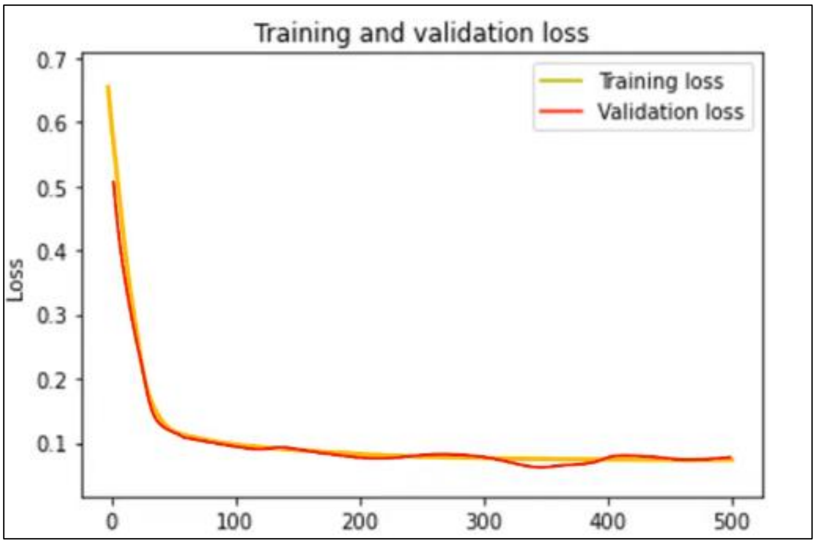

Figure 17 Sample good fit model graph (DigitalSreeni,2020)

Figures 17 show the sample of a good fit model. These two curves converge or come together and validation loss is a bit higher than training loss.

Figure 18 shows the best of this proposed minimizing model loss result. We can see that the curve of loss training and validation is nearly identical, converge and loss training curve is a bit lower than loss validation curve. These curves result shows the best fit model and ready to process recommendation. By watching this graph, we got 12-14 data loss in epoch 1 and then it minimizes data loss until it reaches its point. The best model got at epochs 47- 49.

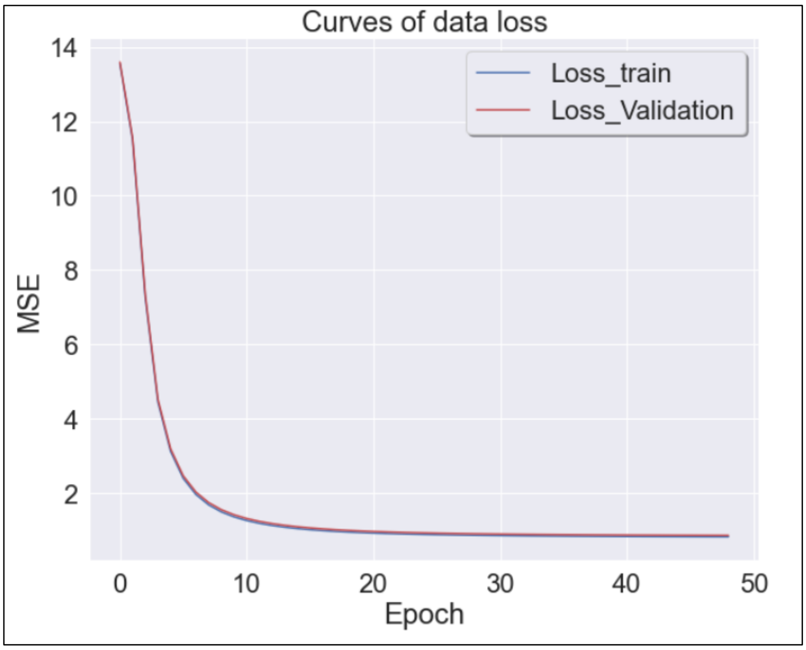

Figure 18 The best model graph after error handling

This is the setting to get this model set to features into 4, learning rate into 0.0001, maximum epochs into 101, regularization or weight_decay into 0.01 and stopping or stop point of minimizing data loss into 0.001.

```
features = 4
lr = 0.001
epochs = 101
weigth_decay = 0.01
stopping = 0.001
```

Figure 19 Setting of error handling

4.4.7. Recommendation

After finishing the error handling, the optimized model or matrix of users and items is received. Therefore, we can recommend using our optimized matrix units for recommending to the users. In this recommendation, this proposed system target is to deliver to each user top 10 recommended items. Additionally, these recommended movies must be unseen or unrated movies for this user.

This function is used for descending order of movies titles and ratings with top numbers. To process this function, it needs titles, ratings and number. It uses numpy.argsort function to sort the data's order with predicted ratings. Thus, the order of the titles and ratings are depending on the numbers of ratings.

```
def top_rank(titles, ratings, n=10):


    # rank indices in descending order
    desc_ranks = np.argsort(ratings)[::-1]

    return titles[desc_ranks][:n], ratings[desc_ranks][:n]
```

Figure 20 Sorting predicted results function

The purpose of this code is for processing of presentation stage or user interface and also recommendation stage. The recommendation target is to recommend to user with the movies that haven't seen by this user.

The optimized model can be used into two options such as user matrix based (specific user) or item matrix based (specific item). In user matrix based, the system takes the unique information of user likes user id to retrieve data from rating record and then calculate by using this user's preference or rating record to recommend the movies for this user. In item matrix based, the system also takes item id to know what kinds of users can like this movie by using ratings records. In addition, we can who is the target of this movie.

This recommendation will perform with user matrix based. Therefore, take user id initially. By using this user id, it will retrieve rating record from the first set of train which has training set with numpy.array to be able to perform matrix factorization. To retrieve data from the data set and set for the unseen movies array, use numpy.where with condition if user specific data equals 0, replace number 1 and otherwise replace 0. So, the movies he hasn't seen or rating become into 1 and his rated movies will be 0 in new unseen rating array. Then seen movie place will show as 0 and unseen movie will show as 1. Another step is matrix factorization results of all movies ratings for this user. Therefore, we can get his or her unseen recommended movies rating by multiplication of matrix factorization results and new rating unseen array. After getting these rating, we will sort ratings and movies with top_rank function into descending order. These are all of the step of the movies.

On user interface, it will request User ID and the numbers of top movies to perform these recommendation result.

```
user_id = int(input("Enter User ID (1-6040):"))

topnumber = int(input("\nEnter Top Number movies:"))

user_train = np.array(train[0][user_id])

unseenmovies = np.where(user_train == 0, 1, 0)

MFPredict= np.dot(P[user_id, :], Q.T)

unseen_movie_predict = MFPredict * unseenmovies

recommendations, prediction_scores = top_rank(np.array(moviesdata['Titles']), unseen_movie_predict, k=topnumber)
print("\nTop",topnumber," highest rating movies for User_ID","[",user_id,"]")
top_n_recommended_data = pd.DataFrame(np.matrix((recommendations, prediction_scores)).T, (np.arange(topnumber) + 1).tolist(
                  columns=['Titles', 'Predicted rating'])

top_n_recommended_data
```

Figure 21 Recommendation stage

We finished methodologies or implementation stage. Therefore, the system can recommend to each user with personalization. We will analyze the systems results and performance of this proposed model in the next chapter.

Chapter 5

Chapter 5 contains evaluation and results of the systems and discussion or critical analysis of these results and evaluation.

5. Evaluation, Result and discussion

5.1. Evaluation

It is impossible to get the perfect model's performance. Thus, most of the model training performances are measured with many Machine Learning evaluation methods. To measure this model performance, regression models (MSE, RMSE and MAE) which are popular in measuring recommendation models have selected. For evaluation result, it takes user matrix and item matrix which have already optimized or minimized loss by error handling and test data which is 20% of the main dataset, which is explained from section 4.4.2 data splitting. This table shows the evaluation methods and its result. These results show approximate values of two decimal points because beyond two decimal places value can change depends on the dataset splitting. This dataset is split with python function and not with manual data split so that the testing and training dataset can change a little bit each time we run.

Table 3 Evaluation Results

Evaluation Method	Approximate value of deviation
MSE	0.86
RMSE	0.92
MAE	0.73

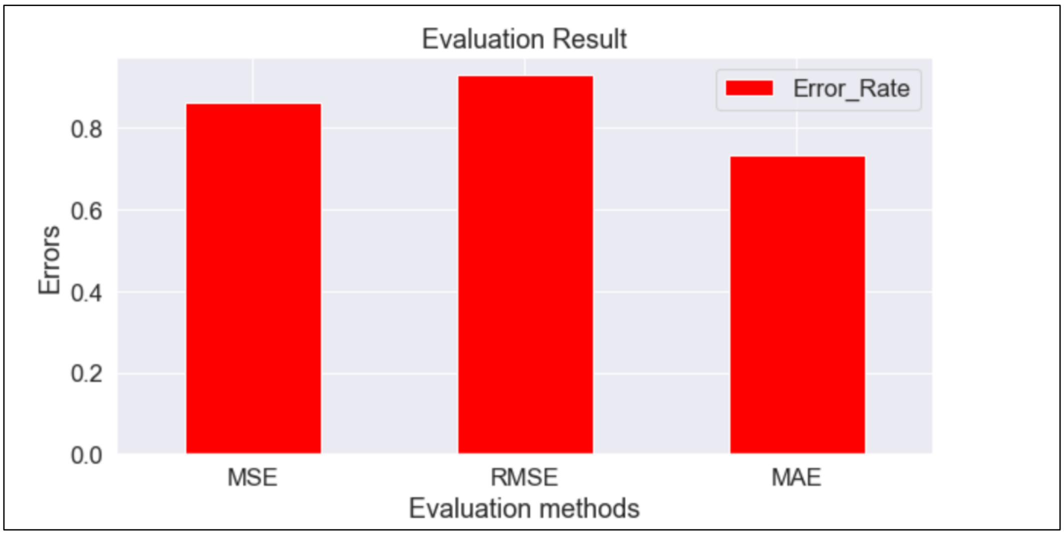

Figure 22 Evaluation Results with graph

5.2. Result

Recommendation system, which was implemented with model (Matrix Factorization) and SGD model, which was well delivered results to the users. The Figure 23 shows the rating records data of user_id 6. The results include user preference's movies titles and predicted ratings.

Top 5 highest rating movies for User_ID [6]

	Titles	Predicted rating
1	Crimes and Misdemeanors (1989)	5.160722
2	Up in Smoke (1978)	5.107708
3	Stuart Little (1999)	5.076507
4	Natural, The (1984)	5.071641
5	Peter Pan (1953)	5.069669

Figure 23 Recommendation results of user id 6

5.3. Discussion

Every system cannot be perfect. Even if this system achieves all goals of project aims, it can still have weak and lack points. After getting results of systems and evaluation, it is important to find the benefits and limitations of this proposed system. If it has limitations, reducing technique for these limitations must add to the future work to be able to upgrade this proposed system in the future.

Prediction accuracy got around MSE 0.8, RMSE 0.9 and MAE 0.7. This deviation result shows that this proposed system performance is quite good. This result remarks us that this proposed model is better than some of traditional collaborative systems. Even though it has to deal with 1M datasets, it can perform well.

This proposed system result is achieved one of the aims of the project which is to deliver the recommendation result of each user with his or her personalization. Because this system uses each user ratings to predict ratings. For a suggestion of this result, only the products should be shown on user's screen which got rating 4 or more is better than showing predicted rating if this proposed system has to combine with video streaming or products sale system. This makes better user-experience and more convenience to user.

Strength of model results are process with optimized model, suggest movies with personalization and predicting accurately. One of the data problems that are commonly found in other recommender systems is data-loss. To overcome this, the performance of SGD and loss function MSE will produce better model which is mention in section 4.4.6. Thus, this proposed system model does not have to worry of data loss. This proposed system processes with each user's rating so that the recommended result can be identical with each user's preference. On the other hand, this proposed system does not have to worry about accurate data. Even if it must deal with large dataset, it's still getting better accuracy result by using enhanced model which got from error handling's performance that is explained in section 4.4.6.

Weakness of model results are cold start problems, UI, data sparsity and time consuming. All of these recommendation results are retrieved from past data of users. However, it provides only for past users or recorded users. So, new users cannot have recommendation results for them. Not only new users but also new items can be one of the limitations. If we add a new movie in this dataset, this new movie cannot be visible on the recommended data because no one engage and make an action on this new movie and this model progresses mainly on the rating data. This output result shows only text format with pandas. So, it has no pictures of movies and no well-designed user interface. This makes user to be bad experience. For developer, this output does not affect obviously to his or her perspective. However, that kind of UI cannot be provide a good experience for normal users. Data sparsity is not too different by comparing with cold start problems. If user has a few numbers of ratings in his or her record, it is hard to predict for this user. Even though it got predicted results, it cannot be accurate. Fortunately, the proposed system's dataset contains at least 20 ratings for each user. If the dataset is changed which has not enough user ratings, this proposed system will be caused problem in predicting. Moreover, according to the cleaned data section of ratings and movies data, extra movies data have to add to perform model if it needs. Thus, data missing or not enough data can cause problems in model demonstration. As an experiment result, error handling stage takes a short time at dealing with a small dataset likes ML 100k datasets. However, it takes a little bit long when the model deals with a large dataset likes ML 1M datasets since it minimizes error from every user's item by using with large numbers of epochs or complete-pass sets. Hence, the larger dataset the model deals, the longer times it will take.

We both learned about the condition of the results and model. The results of proposed system still have lacks. In next chapter, we will study about how to overcome this weakness to improve this proposed system in the future.

Chapter 6

Chapter 6 contains the plans for conquering the weakness or to be improved in the future and conclusion of this thesis.

6.1. Future Work

We learned that in previous section about the weakness of the system. To get better model in the future, we need to overcome or conquer weakness. Firstly, this system does not have User Interface. So, it needs to implement well-designed user interface for providing better user experience.

One of the weakness points of model-based recommender system is cold start problem or new users or items problem. To reduce this problem, content-based recommender system which is one of the most powerful on dealing with implicit data. Some of collaborative filtering systems such as Nearest Neighbor or K Nearest Neighbor can also help in reducing cold start problems. Additionally, we can also reduce by implementing Alternating Least Square. Alternating Least Square has a good ability to deal with implicit data. Additionally, this proposed system is implemented with vanilla Matrix Factorization or standard Matrix Factorization. This matrix factorization We need to upgrade our matrix factorization models. According to JamesLe et al. (2019) explanation, there are seven variations of Matrix Factorization models. This proposed system will be upgraded to Matrix Factorization with side features among seven models. Matrix Factorization with side features is also an improved version of Matrix Factorization with bias. Matrix Factorization with bias can deal with data sparsity by replacing with bias in space of missing values. Side features mean user's demographic and implicit feedback. So that Matrix Factorization with side feature can handle cold start problems and data sparsity problems.

In section 5.3. weakness paragraph, it has a time issue while dealing large datasets. This minimizing model reduces the loss of training and validation as possible as it can so that this minimizing can take long duration depends on the size of the data. We still do not need while model is dealing with 1M or 100K datasets. But its performance will be obviously delayed while model is dealing with 10M or larger datasets. So that we need to improve our error handling model. Omer Faruk Aktulum (2019) demonstrated matrix factorization with upgrade version of SGD model named DSGD (Distributed Stochastic Gradient Decent). That Matrix Factorization can handle even Movie Lens Latest which has 26M ratings. This means that this DSGD can be better than traditional SGD. Thus, DSGD or parallel will be applied instead of SGD to handle larger datasets.

Even though this system can deal with 1M, this is not enough in the real world. Because 1M datasets contains only thousands of movies and users with 1M ratings. Nowadays, movies are increasing day by day and available movies are more than millions. In addition, there are more than six billion people are living in the world and every person has at least one smart device and internet are flexible to use. So that ratings can be more than trillions. By comparing this, our proposed system performance is still quite lack. Not only this system but also all of the machine learning recommender systems cannot perform well. So, we will need better recommender system to be able to perform in real world. To overcome this problem or to construct the most powerful recommender system, deep learning model becomes the most important options.

Deep learning model is state of the art and the most powerful model which process with neural network in recommendation. According to the literature review, deep learning model accuracy is much better than machine learning. So that we do not have to worry on handling large datasets. The more accurate movies delivered, the more satisfaction and enjoy for the users. MF model can be also improved to deep learning model by combining neural network likes RNN (Recurrent Neural Network). If old proposed system does not want to be improved, we can also construct new model of deep learning recommender systems likes Autoencoder, Restricted Boltzmann, Convolution Neural Network (CNN) or neural attention.

Among them autoencoder will be demonstrated to get better performance. Autoencoder, an unsupervised learning for neural network, train the network to ignore noisy signal to learn efficient data compress and encoding or data representations (Deep AI. n.d.). Unsupervised learning is good at handling implicit feedbacks so that one of the one weak points of proposed system has overcome. Additionally, it is also self-supervised for the reason that it can generate own label by itself from the data of training. Therefore, it can also deal with explicit feedbacks. It also has an ability to help in reducing data noise. Noisy data means a data which contains a huge amount of extra meaningless information. In larger movies lens datasets, the data contains loads of information like links, tags, and genome. Not only this information, the other kinds data like implicit feedback from users. Thus, model must have to deal with loads of information to perform. That is why this auto encoder is a good choice for future work. On the other hand, autoencoder outperforms 11 times better than machine learning model where we studied in Section 3. To be able process with deep learning, high end processor or CPU is required. Before demonstration of deep learning start, high performance processor must be had. This is all of the plans of the future for upgrading this old proposed system to high-performance system.

6.2. Conclusion

This thesis target for movie recommender system with personalization. Recommendation systems are used to predict the user's preference by using his or her past records and suggest appropriate movies. To build machine learning recommenders system, there are two types recommendation systems such as content-based and collaborative filtering. This thesis goal is to recommend with personalization this means it must be explicit feedback recommendation. Content-based filtering can powerfully deal with implicit but not with explicit. Most of collaborative filtering system is good at operating with explicit. So, Matrix Factorization algorithm, model-based of collaborative filtering, has better prediction accuracy among collaborative filtering, has selected as a main algorithm. This algorithm becomes main goal of our proposed system. To achieve one of the goals of this thesis which is about to study about Matrix Factorization, we perform 1st aim objectives and these are described in Chapter 3 and Chapter 4 (Section 4.1). For selection of error handling model, SGD is more suitable in explicit than ALS which is covered from existing works in Chapter 3. In addition, most of MF model implementation used the minimizing model of SGD. Hence, SGD model is selected for error reducing. For Matrix Factorization model selection, we chose vanilla MF or standard MF which is easier to construct and have loads of standard MF existing works are available on the Internet.

The 2nd goal is to demonstration of matrix factorization. Datasets were not produced by ourselves so that we download the raw data (Movies Lens) directly from GroupLens. After that, we cleaned it to be able to train MF model. For learning MF model, the loss function (MSE), SGD and MF are used. After improved model was got, we tried to deliver the unseen predicted movies to each user. After we got the results, we measured our model's performance with evaluation methods RMSE, MSE and MAE. We accomplished all of our objectives to achieve our proposed system aims. However, it is not enough with this work done. To get an adequate proposed system, it needs to be analyzed. So, discussion of results that we got is described in Chapter 5. Both of benefits and limitations are found in this system result. To enhance this proposed system, we set future plans in Chapter 6 (Section 6.1). Our plans include Deep Learning (Auto encoder), improved MF (with side features) and other ways to make less or remove limitations. The last but not least, these are all of information that are described in the thesis. We learned that how this MF can carry out 1M datasets proficiently to recommend movies.

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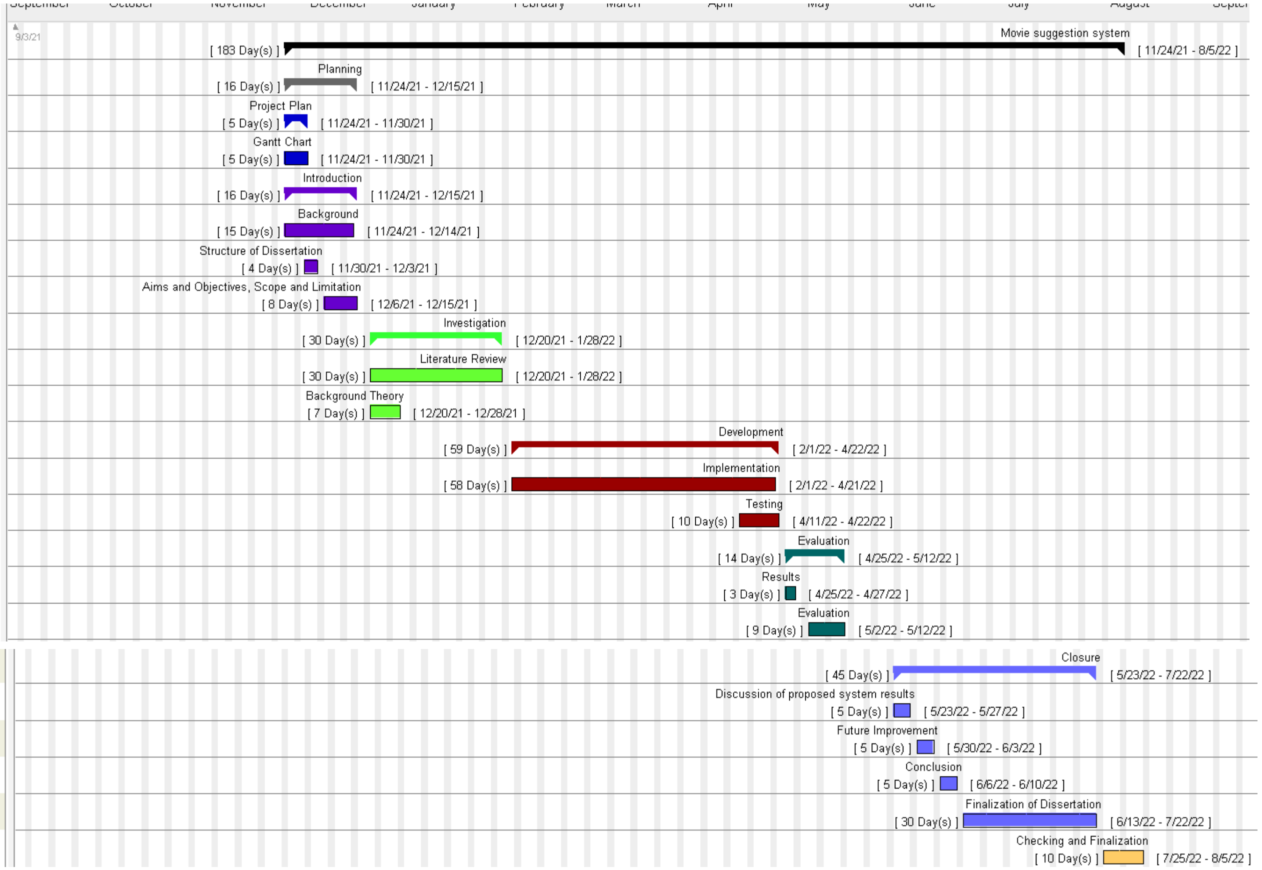
Appendix

Appendix 1

Gantt Chart

Name	Begin date	End date
Movie suggestion syste...	11/24/21	8/5/22
Planning	11/24/21	12/15/21
Project Plan	11/24/21	11/30/21
Gantt Chart	11/24/21	11/30/21
Introduction	11/24/21	12/15/21
Background	11/24/21	12/14/21
Structure of ...	11/30/21	12/3/21
Aims and O...	12/6/21	12/15/21
Investigation	12/20/21	1/28/22
Literature Review	12/20/21	1/28/22
Background The..	12/20/21	12/28/21
Development	2/1/22	4/22/22
Implementation	2/1/22	4/21/22
Testing	4/11/22	4/22/22
Evaluation	4/25/22	5/12/22
Results	4/25/22	4/27/22
Evaluation	5/2/22	5/12/22
Closure	5/23/22	7/22/22
Discussion of pr...	5/23/22	5/27/22
Future Improve...	5/30/22	6/3/22
Conclusion	6/6/22	6/10/22
Finalization of D..	6/13/22	7/22/22
Checking and Finaliz.	7/25/22	8/5/22



Appendix 2

Initial Project Overview

SOC10101 Honours Project (40 Credits)

Title of Project: Movie Suggestion with Personalization Using Matrix Factorization

Overview of Project Content and Milestones

This project will use one of machine learning algorithm to target for the suggestion of movies. Among Machine Learning algorithms, Matrix Factorization which is from Collaborative filtering will be applied on this project to be advanced. Moreover, this project will be implemented with Python language. Not only implementing the recommendation, we will provide the recommendation accuracy result how our model can perform by using with some of the evaluation methods.

To finish every milestone of this project in the given time, Gantt chart will be applied to ensure this project to be finished completely.

The Main Deliverable(s):

This project's outputs will be delivered high accurate predicted to the users with their personalization in a short time.

The Target Audience for the Deliverable(s):

- Movie crazy or movie lover who is keen on watching movie.
- Busy person who has not enough time to spend on searching movie.
- Newbie who doesn't know too much about movies.

The Work to be undertaken:

- Researching and learning similar movie recommendation system.
- Learning and investigation Matrix Factorization.
- Collecting enough dataset.
- Demonstrating model and evaluate model's performance.
- Making a dissertation of the project.

Additional Information / Knowledge Required:

Required knowledge is Matrix Factorization which apply on my project as main algorithm. Moreover, the second priority is to be advanced at Python programming language. The last priority is evaluation model for measuring the performance of the model.

Information Sources that Provide a Context for the Project:

Ramya Vidiyala. (2020, Oct 2). How-to-build-a-movie-recommendation-system. Retrieved Nov 3, 2021. https://towardsdatascience.com/how-to-build-a-movie-recommendation-system-67e321339109

Gulden Uchyigit, Matthew Y. Ma. (2008). Personalization Techniques And Recommender Systems. 5 Toh Tuck Link, Singapore 596224. World Scientific Publishing Co. Pte. Ltd.

The Importance of the Project:

Many machine learning movie recommendation systems have been demonstrated on the Internet using many methods. In fact, there are not too many algorithms which are demonstrated with Matrix Factorization on the Internet. Additionally, matrix factorization outperforms among machine learning algorithm. That is why this project is applied with Matrix Factorization Algorithm and advanced Python language to get the great suggested movies for each user.

The Key Challenge(s) to be overcome:

- Time management might be one of the challenges. In project duration, not only this project but also the other modules must be work done successfully to get Hons degree. Make sure time usage must be balance between this project and other modules.
- Recommendation systems using this project are more difficult than traditional recommendation systems which are using just SQL and Java. So, it might also be a challenge.
- Nowadays, more than thousands of movies and series are available on the Internet. Thus, Data collection might become a challenge.

Appendix 3

SOC10101 Honours Project (40 Credits)

Week 9 Report

Student Name: Sai Bhone Myat Naing

Supervisor: Dr Aye Aye Myint

Second Marker: Dr Kehinde Babaagba

Date of Meeting: 27/01/2022

Can the student provide evidence of attending supervision meetings by means of project diary sheets or other equivalent mechanism? **yes**

If not, please comment on any reasons presented

Please comment on the progress made so far

The student has made some progress and has provided a literature review, introducing the concepts as well as related work. The student is slightly behind schedule based on the GANTT chart provided but plans on getting back on track in the coming weeks.

Is the progress satisfactory? **yes**

Can the student articulate their aims and objectives? **yes**

If yes then please comment on them, otherwise write down your suggestions.

The student understands their project aims which includes the construction of a Movie Recommender using Matrix Factorization. The objectives were also well articulated by the student.

Does the student have a plan of work? **yes**

If yes then please comment on that plan otherwise write down your suggestions.

The student has a plan of work evidenced by a GANTT chart but is slightly behind schedule by about 10 days. The student should be at the preliminary stages of the implementation of the project based on the GANTT chart but is not there yet. The student plans on getting back on track in the coming weeks.

Does the student know how they are going to evaluate their work? **yes**

If yes then please comment otherwise write down your suggestions.

The student has a bit of understanding of how evaluation would be done for their work and mentioned one metric they will be using. I would advise that for benchmarking purposes, they use a number of metrics backed up by literature and this will help the quality of the work.

Any other recommendations as to the future direction of the project

In general, I would say the student has made some progress and the work represents a good challenge for the student. The student has also stated that they will make effort to get back on track with regards to their project timeline. Other than the already mentioned recommendations, I would advise that the student ensures that all formal document elements are included in the final report, as the report submitted was missing a title page. Also, please check references as some were missing like 5A. Furthermore, in terms of the structure of the literature review, I believe the background theory should come before the related works. This is so that when a reader of the work sees the related work, they already have a background knowledge of the theories used in those papers. Finally, please ensure that you provide more in-depth justification of methods/tools and dataset employed in the project.

Signatures: Supervisor Aye Aye Myint Second Marker KEHINDEBABAAGBA

Student

The student should submit a copy of this form to Moodle immediately after the review meeting; A copy should also appear as an appendix in the final dissertation.

Appendix 4

PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 11/20/2021 **Last diary date:**

Objectives:

to discuss overall project idea, the contents of dissertation of LO1, such as abstract, aim and objectives, problem statement

Progress:

project started from scratch.

Supervisor's Comments:

to write abstract, aims and objective, problem statement etc, gantt chart and workbreakdown structure
to learn papers relating the project scope
to learn machine learning in depth
to learn some recommender system on machine learning system
to install anaconda

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SCHOOL OF COMPUTING

PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 11/27/2021 **Last diary date: 11/20/2021**

Objectives:

to check what is going on with his project

Progress:

wrote abstract , aims and objectives

Supervisor's Comments:

to learn related works by exploring academic papers based on movie recommender system with collaborative filtering
to consider how to evaluate the performance of the recommender system (which method will be used) and update on dissertation

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 12/15/2021 **Last diary date: 11/27/2021**

Objectives:

to check what is going on with his project and how much he has understood his project scope.

Progress:

learnt matrix factorization approach on movie recommendation. learnt machine learning, deep learning and AI. learnt RNN

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 12/15/2021 **Last diary date: 11/27/2021**

Objectives:

to check what is going on with his project and how much he has understood his project scope.

Progress:

learnt matrix factorization approach on movie recommendation.
learnt machine learning, deep learning and AI.

Supervisor's Comments:

can explain matrix factorization approach for movie recommendation.
he will finish LO1 contents .

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 12/15/2021 **Last diary date: 11/27/2021**

Objectives:

to check what is going on with his project and how much he has understood his project scope.

Progress:

wrote aims and objective

Supervisor's Comments:

supervisor reviewed and revise aim and objectives
supervisor explained how to write problem statements
supervisor guide how to do citation and reference in writing literature review

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 1/8/2022 **Last diary date: 12/15/2021**

Objectives:

to be ready for interim meeting

Progress:

has written chapter 1 and partial portion of chapter 2.

Supervisor's Comments:

-suggest how to write project scope
-there is enough progress in exploring literature and has done reviewed well. However, it is not completed.
need to study matrix factorization for the sake of fully understanding

I would suggest which should be included in the scope and limitation.
suggestion for checklist of project scope

what will be explored?
intended user?
intended ouput?
Data Set that will be used ?
algorithm that will be applied?
interface? command line or with GUI

Limitation (what will you not do in the research because of time or something)
you cannot test it by building web site because of time limitation

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 2/7/2022 **Last diary date: 1/8/2022**

Objectives:

To discuss second marker feedback

Progress:

Understand matrix factorization concepts

Supervisor's Comments:

He has learned Matrix factorization. Should start the design and implementation

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 3/3/2022 **Last diary date: 2/7/2022**

Objectives:

To check how the progress is

Progress:

He is trying to prepare dataset.

Supervisor's Comments:

Need to learn from some machine learning textbook
Need to explore github , torrent for getting some suitable dataset.

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 3/17/2022 **Last diary date: 3/3/2022**

Objectives:

To check how the progress is

Progress:

Training data has be defined ad MovieLens dataset

Supervisor's Comments:

Need to train , test alogorithm

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date: 3/25/2022 **Last diary date: 3/17/2022**

Objectives:

To check how the progress is

Progress:

Implementation is in progress

Supervisor's Comments:

need to explore sample code and learn

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date 4/9/2022 **Last diary date: 3/25/2022**

Objectives:

To check how the progress is

Progress:

Implementation is in progress and model training started

Supervisor's Comments:

need to explore sample code and learn

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date 5/6/2022 **Last diary date: 4/6/2022**

Objectives:

To check how the progress is

Progress:

Implementation is almost done and need to explore evaluation metric

Supervisor's Comments:

need to study evaluation metric for recommendation system

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date 5/20/2022 **Last diary date: 5/6/2022**

Objectives:

To check how the progress is

Progress:

He explored evaluation metric such as RMSE, MAE

Supervisor's Comments:

need to integrate performance evaluation metrics code

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PROJECT DIARY

Student: Sai Bhone Myat Naing **Supervisor: Dr. Aye Aye Myint**

Date 6/10/2022 **Last diary date: 5/20/2022**

Objectives:

To check how the progress is

Progress:

Performance evaluation code has been integrated

Supervisor's Comments:

need to complete dissertation book